Denoising Diffusion Models

A short tour of the forward and reverse processes

Outline

1. Motivation
2. The forward process
3. From data to noise
4. Sampling

Part 1. Motivation

Part 2. The forward process

Why diffusion models?

- Generate data by reversing a gradual noising process
- Training is a simple regression loss
- Tractable likelihood bounds and flexible samplers

Part 2. The forward process

Part 3. From data to noise

The forward process

Adding noise, step by step

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$$

Definition

A sequence $\beta_1, \ldots, \beta_T$ sets how much noise each step adds.

```
def q_sample(x0, t, eps):
    a = alpha_bar[t]
    return a.sqrt() * x0 + (1 - a).sqrt() * eps
```

Part 3. From data to noise

Part 4. Sampling

From data to noise

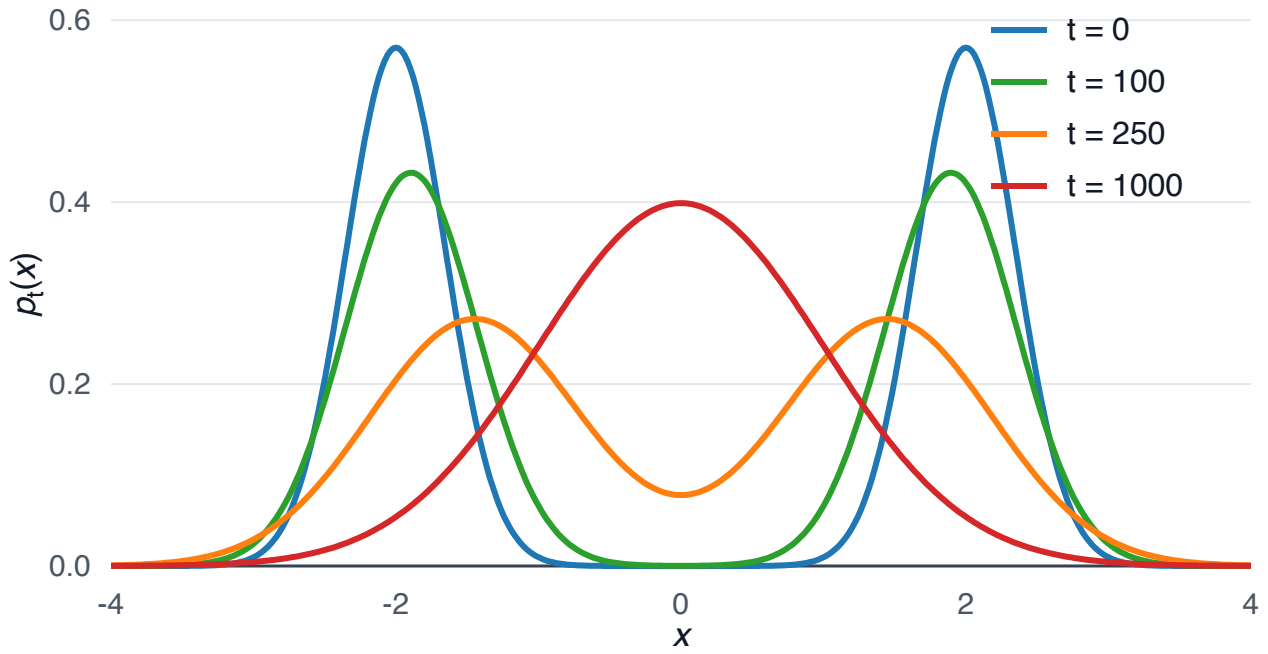


Figure 1. Data is gradually turned into Gaussian noise.

Part 4. Sampling

References

Song, J. et al. (2020). Denoising Diffusion Implicit Models. arXiv:2010.02502.

Thank you